London Ambulance Service NHS Trust

Support the management of trauma in the emergency and urgent care setting
Component 8

Spinal & Pelvic Injuries

Pelvic Injuries - Component 8

Objective

Identify the mechanism of injury, the signs and symptoms and describe the management of pelvic and spinal injuries.

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Pelvic Injuries - Component 8

Pelvic Injuries

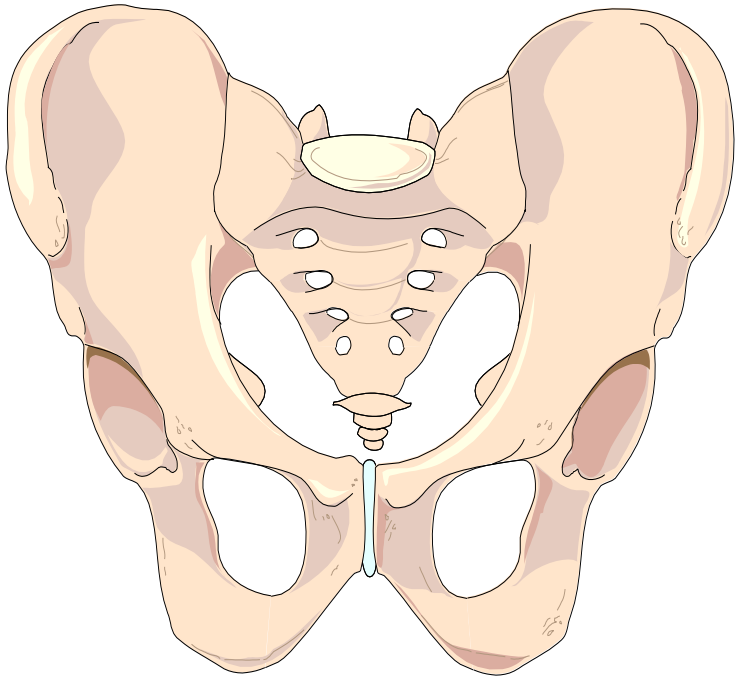
- Pelvic fractures represent 3-6% of all fractures in adults
- Occur in 20% of all polytrauma cases
- 75% of all pelvic injuries occur in men
- Haemorrhage is the cause of death in 40% pelvic trauma patients



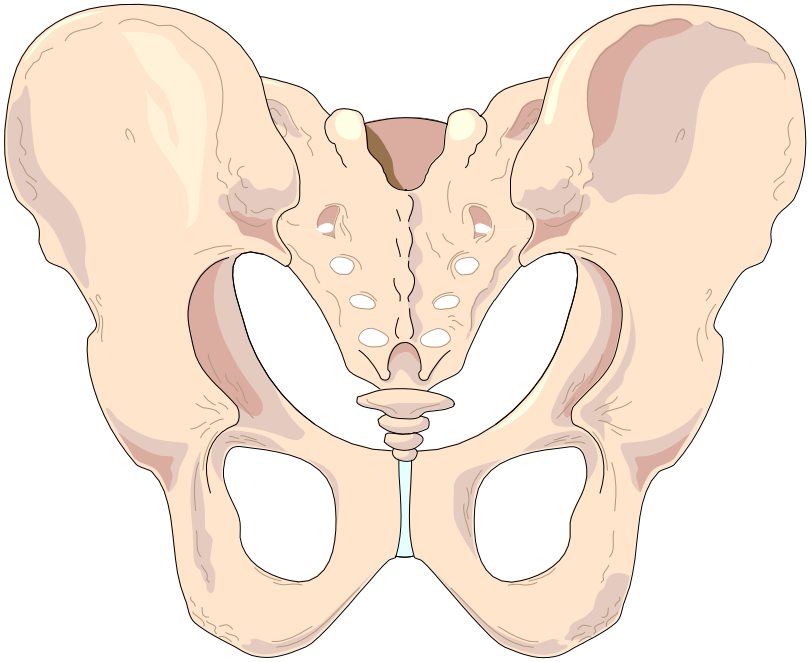
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Pelvic Injuries - Component 8

The Pelvis



Anterior view



Posterior view

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Pelvic Injuries - Component 8

Causes of Pelvic Injuries

- High energy transfer e.g. RTC
- Fall from height
- Crush injury



Can also be from simple falls with risk factors:

- Elderly
- Degenerative bone diseases
- Receiving radiotherapy



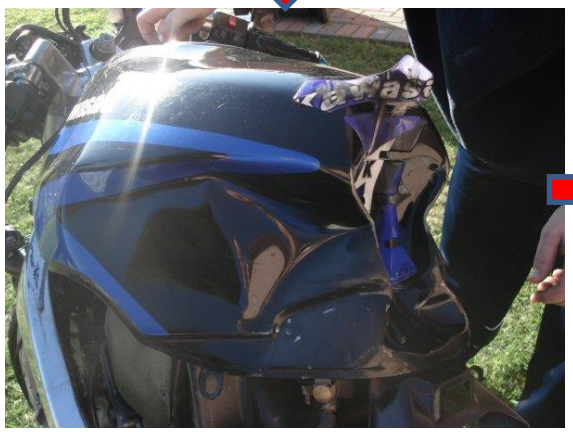
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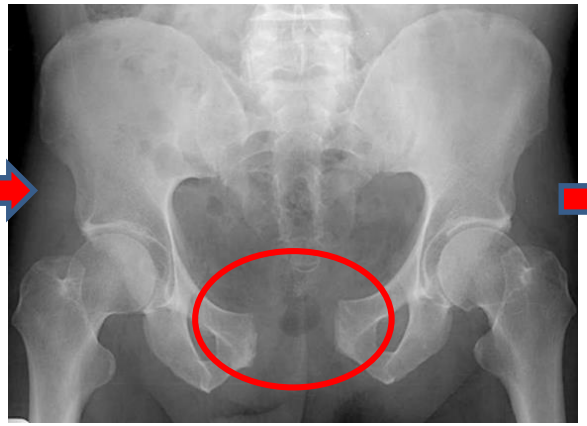
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Pelvic Injuries - RTC



Any significant door intrusion into the passenger space should give cause for concern, as should a dented petrol tank on a motor bike.







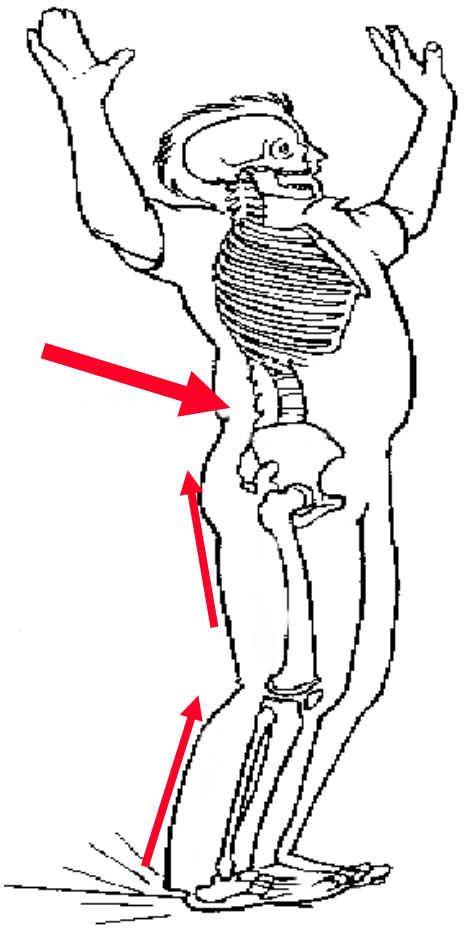
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Pelvic Injuries - Falls from Height

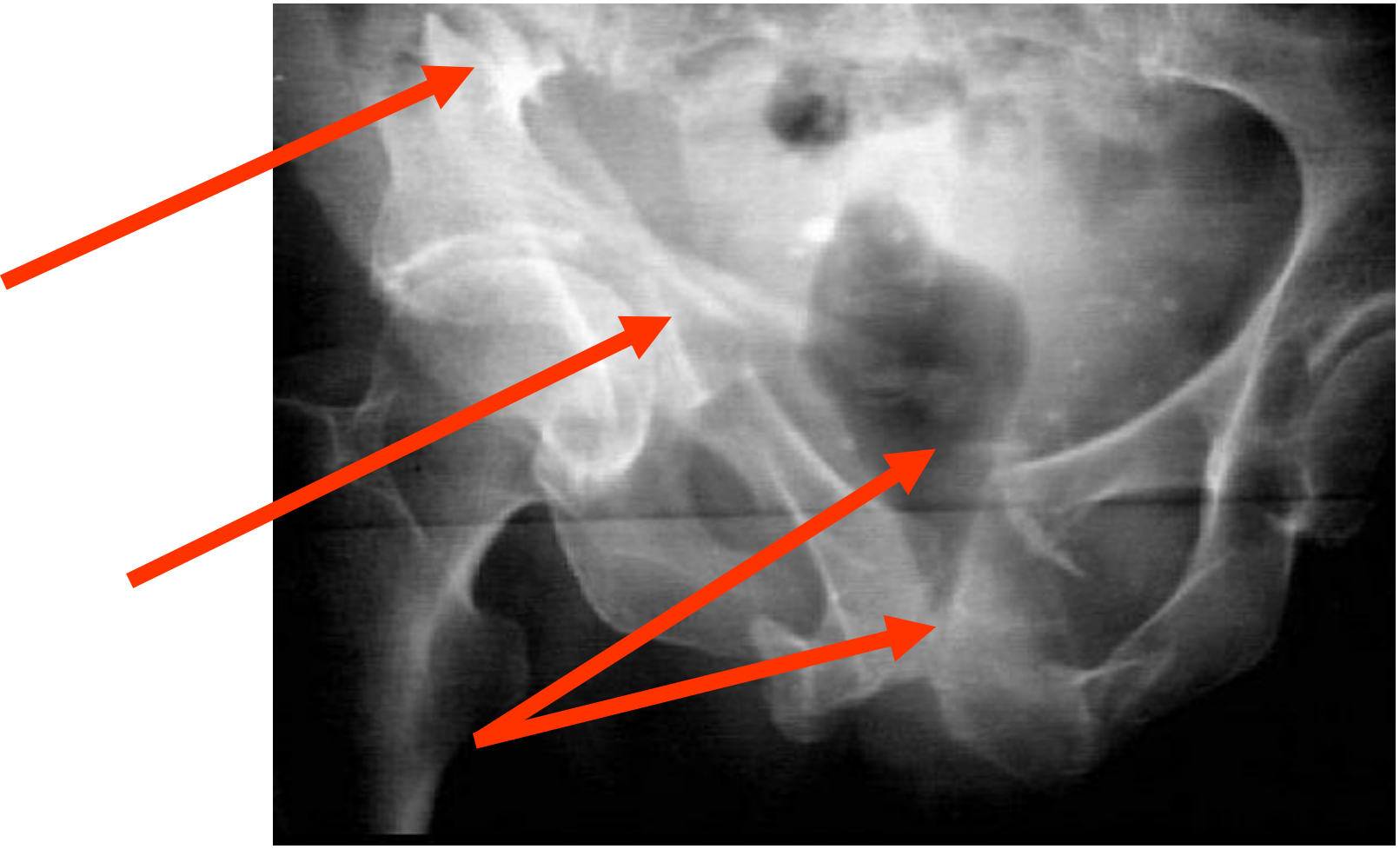
Remember

Distracting injuries can occur
e.g. fractured ankle



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Complications

- Increased pelvic volume due to fracture allows for increased haemorrhage
- Vascular injuries leading to haemorrhage
- Urogenital injuries – bladder rupture in 10%
- Common to have other injuries to manage e.g. intra-thoracic/intra-abdominal

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Signs & Symptoms

- Mechanism
- Bruising
- Bleeding from the patient's rectum, vagina or urethra
- Deformity
- Swelling
- Shortening of lower limb
- Pain over the hips, groin or lower back
- Hypovolemic shock
- Altered sensation in one leg

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Management of Pelvic Injuries

What does the JRCALC Guideline say?



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Pelvic Injuries - Component 8

Management

Time critical – DRABC & Splint



- Consider C-spine when managing airway
- Control catastrophic haemorrhage
- High flow O_2
- Pelvic Splint – immobilise knees and ankles as well
- Consider Paramedic assistance
- Pain relief
- Smooth rapid removal to MTC
- Full observations



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Management

- Avoid repeated examinations/movements
- Never ‘spring’ the pelvis –



  - this may dislodge blood clots
  - may exacerbate injury
- Once the pelvic splint has been placed – do not remove

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Pelvic Splint

- As the pelvic splint is cut to size, it can fit a massive range of patients from paediatric to bariatrics

- It is a single use device



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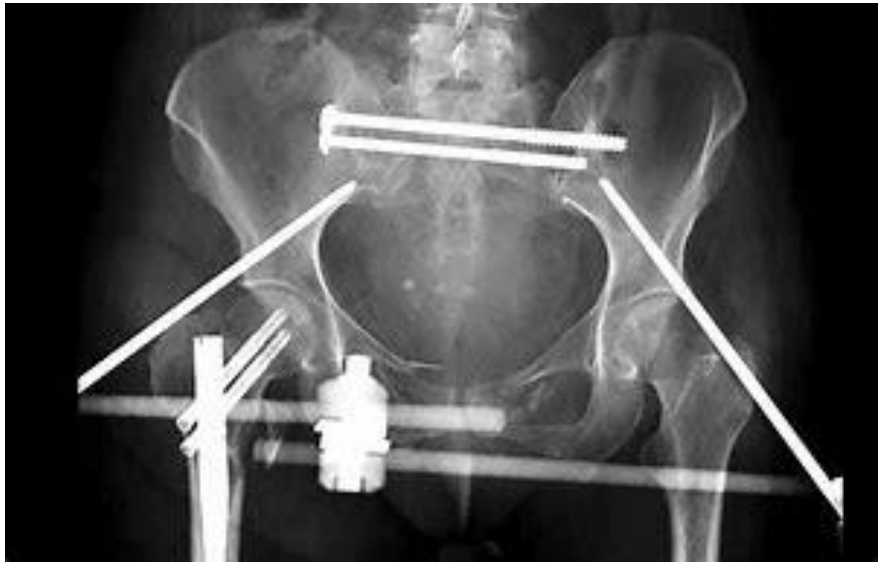
Summary

- Pelvic injuries can lead to fatal haemorrhage
- Minimise movement when treating pelvic injuries and limit log roll to 10 degrees
- Pelvic injuries require immediate splinting and transport to MTC
- Pelvic splints are single-use only and once applied, will not be removed by ambulance clinicians.

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Pelvic Injuries - Component 8

Any Questions?



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Spinal Injuries - Component 8

Spinal Injuries



Spinal Injuries - Component 8

Mechanisms of Injury

- Hyperflexion
- Hyperextension
- Rotation
- Compression
- One or more of these



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Spinal Injuries - Component 8

How Does Spinal Injury occur?

- 46% from **Falls**
  - A third of these are less than 2 metres
  - Increased risk in older people/Rheumatoid Arthritis
- 40% from **Road Traffic Collisions** – what increases the risk factors?
- 3% from **Sporting injuries**

Spinal Injuries - Component 8

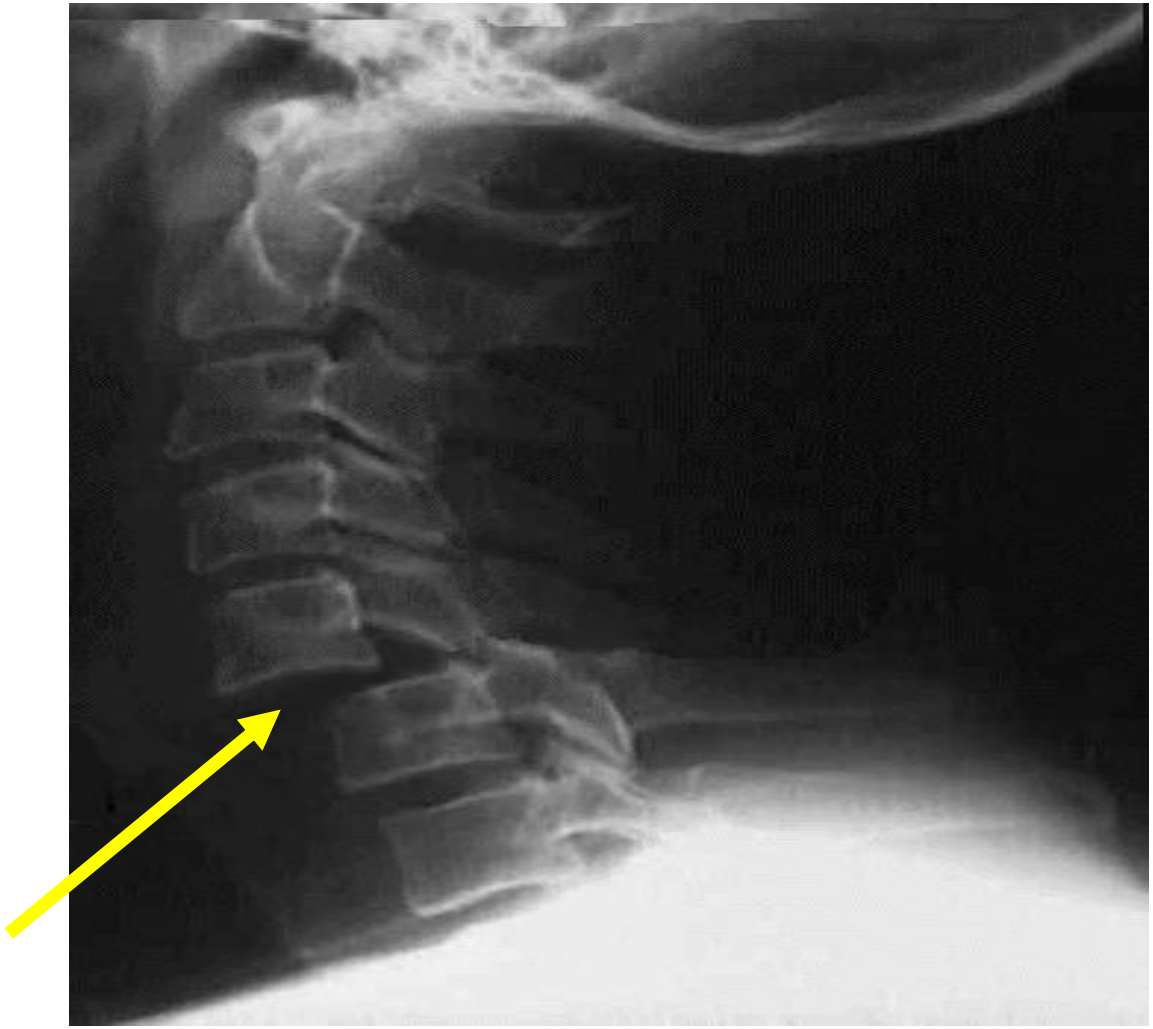
Types of Spinal Injuries

- Dislocation of vertebrae
- Fracture of vertebrae
- Displacement of vertebrae (prolapse)

All of the above can lead to compression and/or stretching of the spinal cord leading to Spinal Cord Injury (SCI)

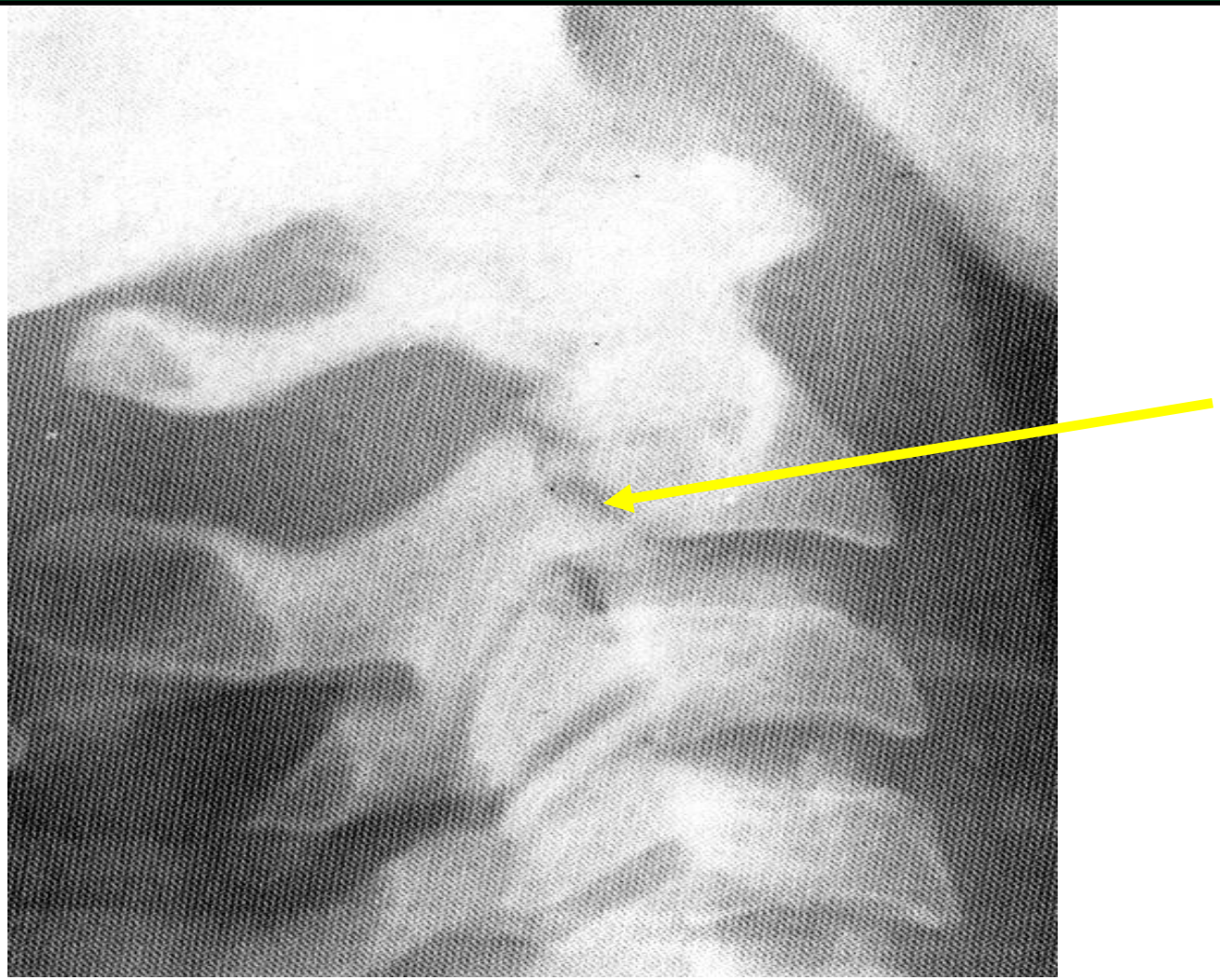
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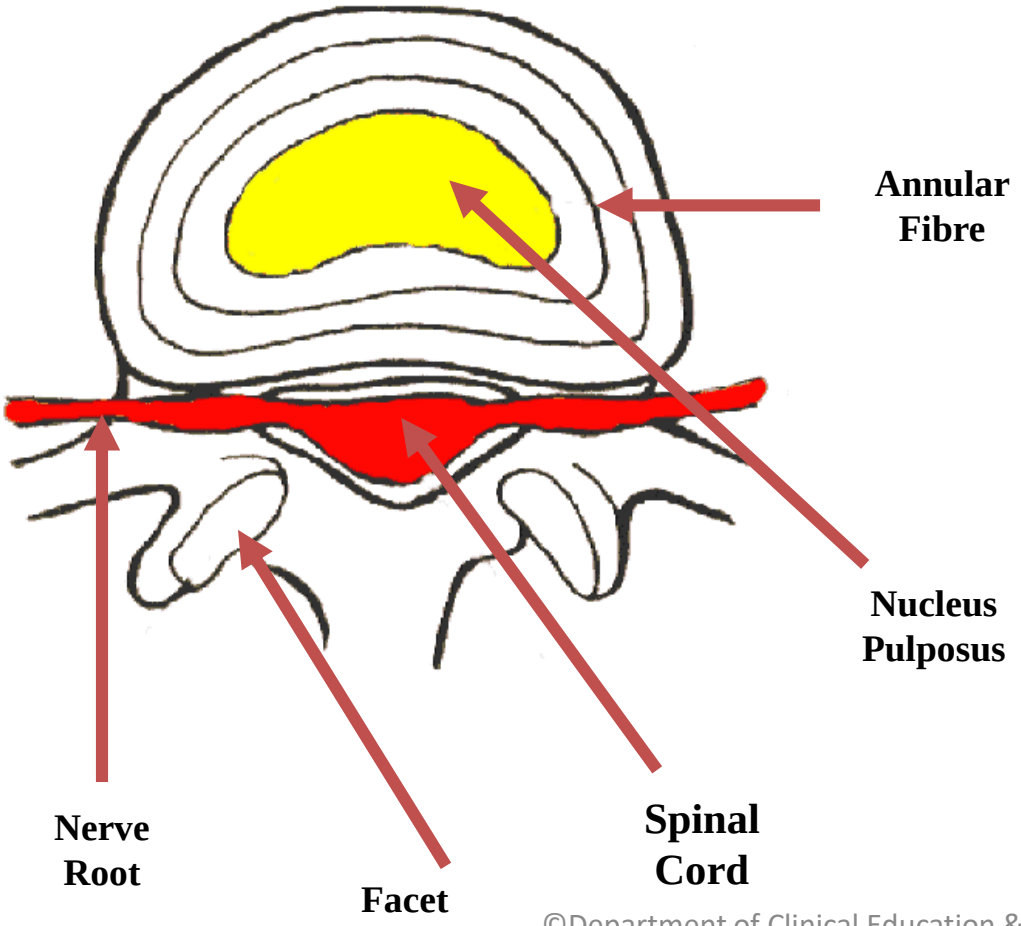
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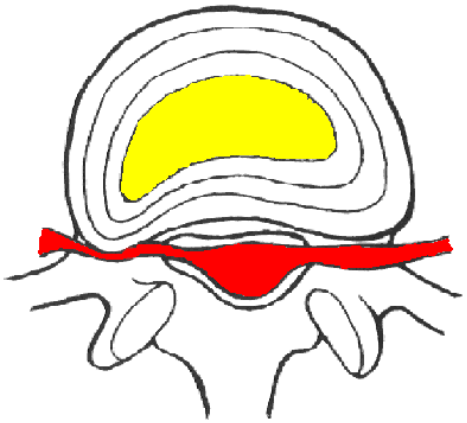
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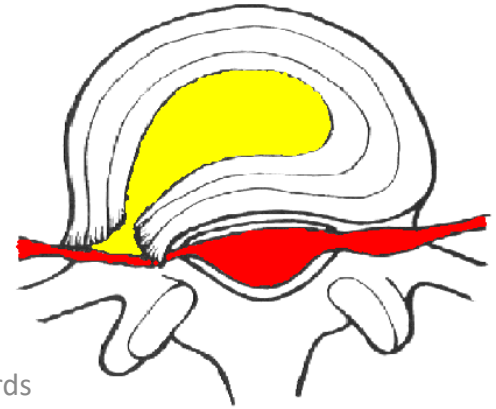
Healthy Disc



Bulging Disc



Herniated Disc



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Spinal Injuries - Component 8

Other causes of Spinal Injuries

- Spinal Tumours
- Vascular thrombosis or haemorrhage
- Infection
- Abscesses caused by TB or meningitis

Spinal Injuries - Component 8

Complications of Spinal Injuries

- Damage to spinal cord
- Compression of spinal cord
- Spinal shock
- Neurogenic shock
- Respiratory difficulties
- Cauda Equina Syndrome

Spinal Injuries - Component 8

Spinal Shock

After Spinal Cord Injury (SCI):

- State of complete loss of motor function
- Can have possible loss of sensory function
- Can go on for considerable time
- Recovery may occur

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Spinal Injuries - Component 8

Neurogenic Shock

After Spinal Cord Injury (SCI):



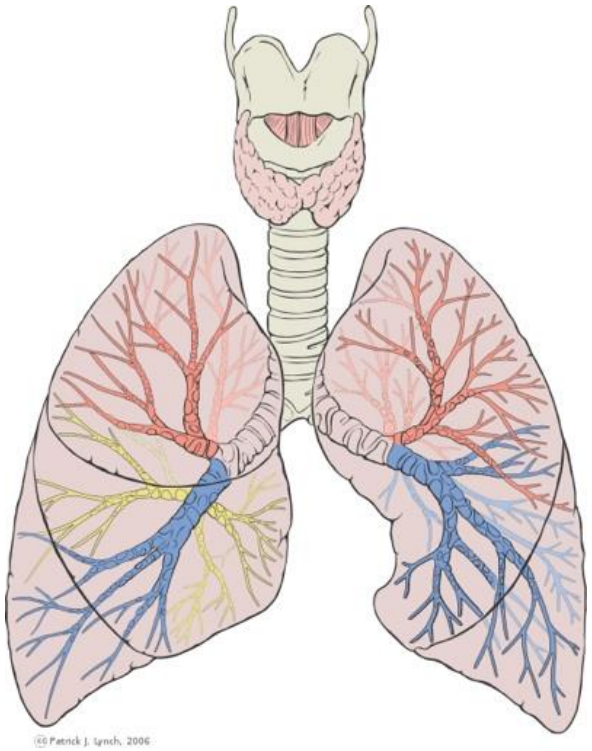
- Poor tissue perfusion
- Caused by sympathetic tone loss

Spinal Injuries - Component 8

Respiratory Difficulties

After Spinal Cord Injury (SCI):

- Damage to upper cervical cord
- Paralysis of diaphragm and respiratory muscles



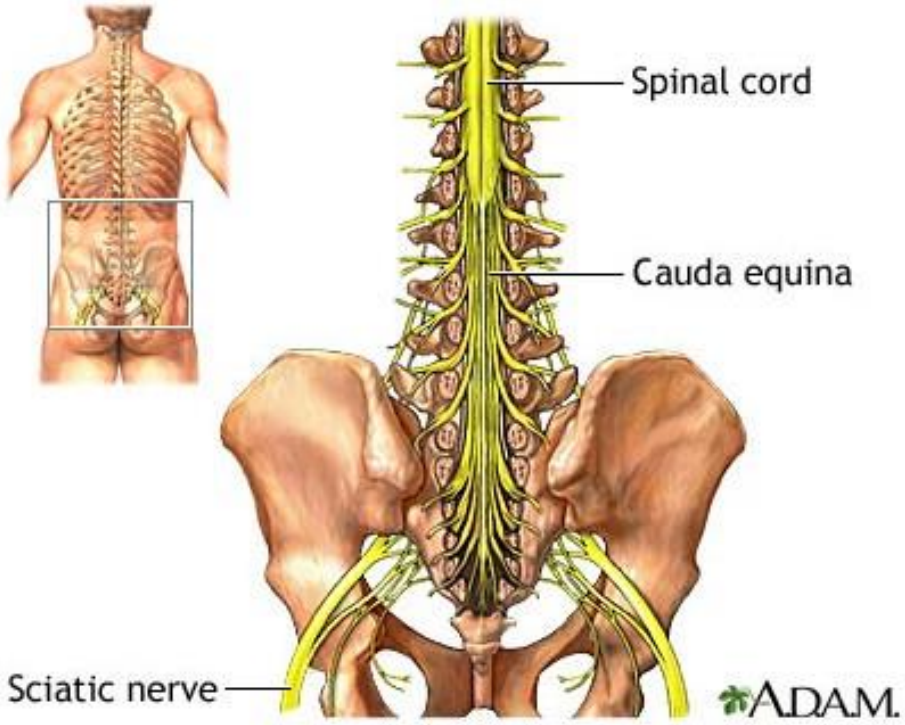
Spinal Injuries - Component 8

Cauda Equina Syndrome

- Severe compression or inflammation of spinal nerves in lowest region of spinal canal
- Onset can be acute (hours/days) or chronic
- It is a surgical emergency – urgent decompression required

Spinal Injuries - Component 8

Cauda Equina Syndrome



Clinical Red Flags:

Any of the following are significant in the presence of lower back pain (generally sharp or stabbing) and/or unilateral or bilateral lower limb pain, radiating from the lower back (sciatic type pain):

- **Numbness in the groin/perineum/buttocks (saddle anesthesia)**
- **Bladder or bowel dysfunction**
- **Sexual dysfunction**
- **Lower limb weakness and/or sensory deficit (disturbed gait/inability to walk)**
- **Reduced or absent lower limb reflexes**

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Cauda Equina Syndrome

Did *you* know?

The NHS Litigation Authority received **293** claims for incidents of Cauda Equina Syndrome occurring between January 2010 to December 2015.

Cauda Equina Syndrome can result from a ruptured disk in the lower back, a narrowing of the spinal canal, a spinal tumour, fracture, hemorrhage or infection. It can also result from a trauma such as a car accident, gunshot wound, a stabbing or a fall.

The patient's vertebral discs prolapse and compress the nerves descending from the end of the spinal cord (the cauda equina).



232

claims are under investigation reflecting the complications of the injury and long term consequences for the patient.

20

cases with damages paid

41

cases resolved with no damages paid

This has cost the NHS

£25,200,000

this includes damages, defence and claimant costs.

Key causes are due to delays in:

- specialist centre referral;
- diagnosing/recognising symptoms;
- MRI scanning (time, access, referral);
- responding to 'red flag' symptoms;
- surgical intervention; and
- transport to hospital or specialist centre.

It may also be as a result of surgery.

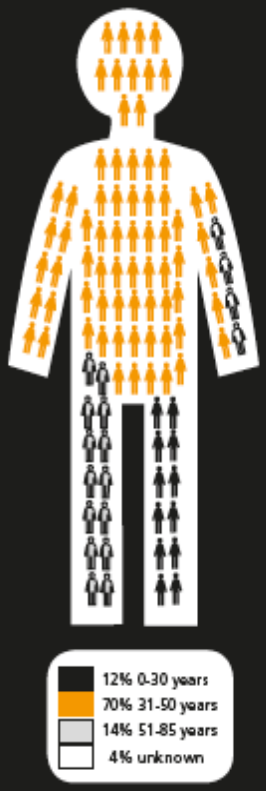
If the condition is not addressed at an early stage, it can result life-changing injuries which are often associated with significant psychological effects.

Cauda Equina Syndrome primary symptoms include:

- bladder/bowel dysfunction;
- loss of sexual function/impotence;
- nerve damage;
- unnecessary pain;
- paralysis/partial paralysis/disability/ foot-drop; and
- spinal damage.

In some cases, Cauda Equina Syndrome can contribute to the risk of death.

Age range of patients



'Red flag' symptoms that require rapid investigation and treatment can include:

- anal and/or buttock numbness;
- bladder or bowel disturbance;
- loss of feeling between the legs (saddle anaesthesia);
- severe back pain; and
- loss of sexual function.



Spinal Injuries - Component 8

Signs & Symptoms of Spinal Injury

Common Signs/Symptoms of fracture and/or:

- Pain anywhere along the spine (neck or back)
- Loss of sensation in the limbs
- Loss of movement in the limbs
- Altered sensation in the trunk or limbs (of pins and needles/ burning/ electric shock)



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Signs & Symptoms of Spinal Injury

Associated symptoms may indicate level of injury:

- Hypotension <80mmHg with bradycardia in C-spine injuries
- Diaphragmatic or abnormal breathing
- Hypotension with warm peripheries or vasodilation
- Flaccid muscles with absent reflexes
- Priapism

Spinal Injuries - Component 8

Management of Spinal Injury

- Safety
- <C> ABCDE
- Maintain c-spine at all times, with cervical collar or manual in-line stabilisation (MILS)
- Extricate patient to a safe place for assessment
- Asking a patient to self extricate is acceptable providing that……..
- Assess and manage pain
- Assess the patient using immobilisation algorithm



Spinal Injuries - Component 8

Management of Spinal Injury

- Immobilise entire spine if indicated (collar, blocks & scoop)
- Rescue/spinal boards are for ***extrication*** only
- If immobilisation is not possible, use padding/support
- Use major trauma triage tool to determine conveyance
- Pre-alert the receiving hospital if necessary
- If time permits, carry out secondary survey
- Convey smoothly



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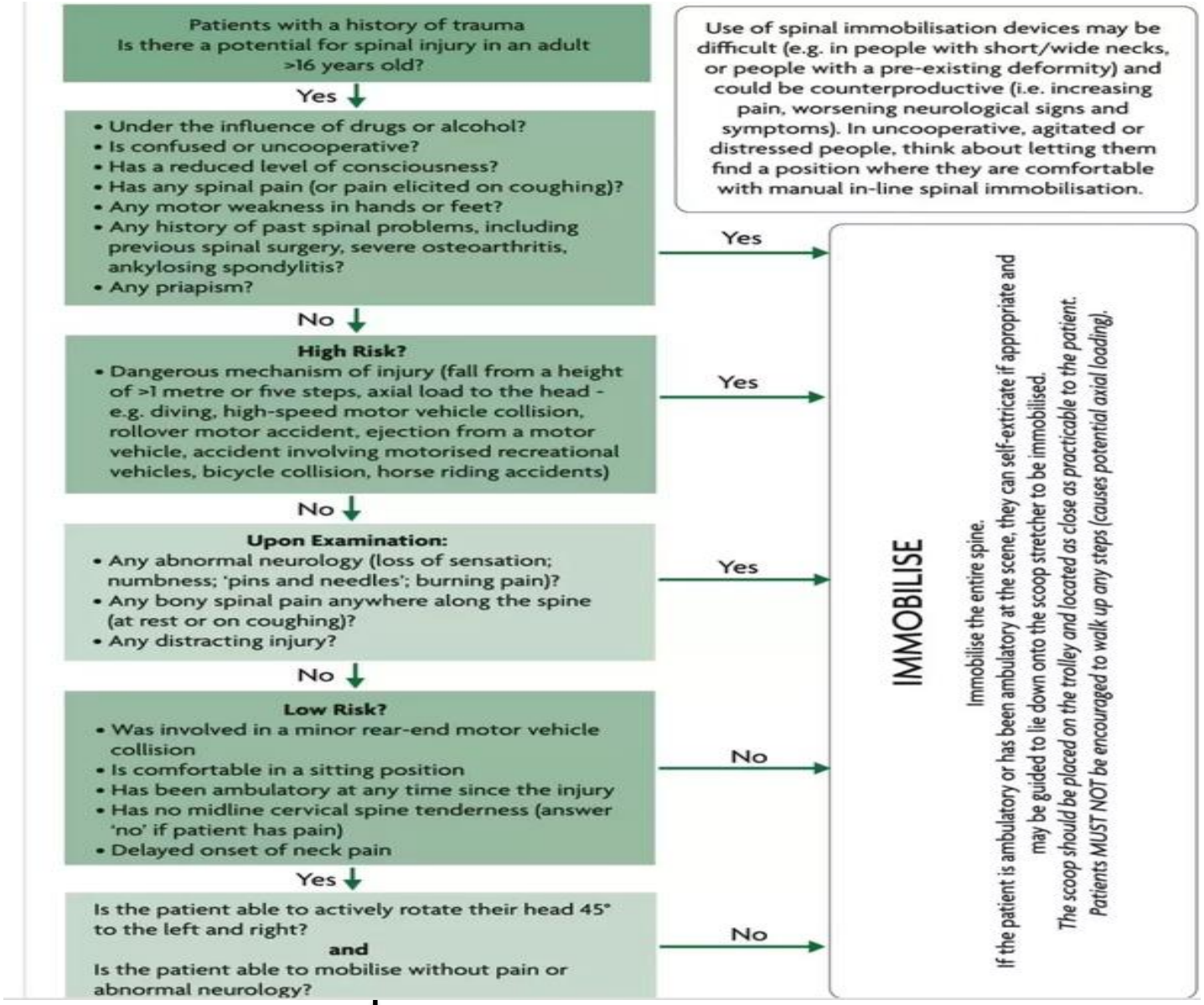
Spinal Injuries - Component 8

Management of Spinal Injury

Potential complications when managing the patient:

- Airway problems – increased risk of aspiration
- Increased intracranial pressure
- Restricted respiration
- Dysphagia
- Skin ulceration/pressure sores
- Pain

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Spine Cleared

Patients with suspected spinal injury with Abnormal neurology must be transferred To a Major Trauma Centre

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Spinal Injuries - Component 8

Reminder to complete **full documentation**.

- Was MILS used to maintain c-spine?
- How was the patient extricated?
- What were the clinical findings from using the immobilisation algorithm?
- Was the entire spine assessed, or just the c-spine?
- Was the patient immobilised? If so, with what equipment?
- Which step of the Major Trauma Tool was triggered, if any?

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Spinal Injuries - Component 8

Summary

- Any doubt – treat for a spinal injury
- Keep control of c-spine until full immobilisation has been carried out
- Utilise JRCALC immobilisation algorithm
- Immobilise whole spine until positively cleared
- Handle patient carefully
- Transport smoothly



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